

Code Structure — Intelligent University Course Finder

Project Root

```
Capstone_Project/
|
├─ README.md           # Main project documentation
├─ README.pdf          # PDF version of README
├─ CODE_STRUCTURE.md   # This file
├─ CODE_STRUCTURE.pdf  # PDF version of CODE_STRUCTURE
├─ design_decisions.md # Technical design rationale
├─ design_decisions.pdf # PDF version of design_decisions
├─ .gitignore          # Git ignore rules
|
├─ Architecture_Diagram.png # System architecture visualization
├─ Data_Flow.png        # AI Agent data flow visualization
├─ Data_Ingestion.png   # Data processing pipeline visualization
├─ Search_Flow.png      # Hybrid retrieval flow visualization
|
├─ Source/              # ALL SOURCE CODE LOCATED HERE
|   │
|   ├─ .env              # Environment variables (OPENAI_API_KEY, etc.)
|   ├─ config.py         # Centralised configuration
|   ├─ logger.py         # Logger factory
|   └─ requirements.txt   # Python dependencies
|   │
|   ├─ api/              # Backend microservice (port 8000)
|   │   │
|   │   ├─ main.py       # FastAPI entry point
|   │   └─ schemas.py    # Pydantic models
|   │       └─ core/      # recommend_logic.py (Orchestrator)
|   │
|   ├─ api_gateway/      # API Gateway (port 8080)
|   │   └─ gateway.py    # FastAPI proxy
|   │
|   ├─ app_agents/       # AI Agents (OpenAI SDK)
|   │   │
|   │   ├─ intent_agent.py
|   │   ├─ skill_gap_agent.py
|   │   ├─ career_agent.py
|   │   ├─ sequencer.py
|   │   ├─ advisor_agent.py
|   │   └─ tools.py      # Agent function tools
|   │
|   ├─ retrieval/        # Hybrid Search logic
|   │   └─ hybrid_retriever.py
|   │
|   ├─ guardrails/       # Safety & Validation
|   │   └─ validator.py  # Presidio + intent check
|   │
|   └─ ingestion/        # Data ingestion pipeline
|       └─ ingest.py
|   │
```

